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AI-Edge REU Internship

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During this two-month period of participating in the AI-EDGE summer REU program, I feel I've learned a lot about the complexities of machine learning, and what is really occurring underneath the flashy AI programs that are rapidly improving in capability and popularity, such as ChatGPT and DALL-E. We began the program with an overall introduction into machine learning, with its mathematical foundations and an understanding of how it actually "learns." Training AI to interpret and sort fashion items via the FASHION-MNIST dataset was not only an engaging way to start the program, but also helped and reinforced my growing understanding of Python. I was able to use it to run python code that trains the AI on this dataset, and I observed how the parameters such as learning rate contribute to its ability to get better at distinguishing different clothing articles. This was also where I was actually able to observe training loss values, how they change over time and contribute to the loss curve of a training session. Moreover, I also learned about transformers and how Language Models use data to improve. This required a review of probability concepts I've learned in university courses I've taken, such as Taylor's approximation and gradient descent, but reinforced these concepts in their relation to Transformers and Attention. In Gradient Descent, I learned that there is a tradeoff when tweaking the learning rate. A larger learning rate will result in fast convergence but high error, while smaller learning rates results in slow convergence but lower residual error. Another concept that especially interested me was Attention, which *intuitively* is the contextual embedding of words in a sequence based on information from its neighboring words, and

mathematically is a way to take a weighted sum of vectors. These are fundamental concepts in machine learning that helped me understand other concepts such as tokens and embeddings, as well as LLMs as a whole. Not only did these few weeks highlight how “large” these popular language models really are, but also ironically how many small algorithms and components are actually needed to help them function. In the latter half of the REU we explored different applications of AI, which was interesting as it showed how versatile and impactful AI can be as it becomes more and more integrated in our everyday lives. I learned about the basics of wireless networks such as 5G, and the concept of handover in mobile devices. AI can essentially target the challenge that we face with handover in 5G Networks, as it will be able to use behavioral patterns to predict where users are moving, and therefore maintain network functionality as the mobile device moves from place to place. We also dived into AI for security design, which is important as network security and cybersecurity attacks in general continues to be a prevalent threat today. I learned the “nut’s and bolt’s” of the internet, how network protocols allow communication and requests, which was both a refresher and an expansion of concepts I learned in my time as a Network Engineer Intern with UIC Tech Solutions. Overall, I’m very grateful to AI-EDGE and Ohio State for the ability to participate in this summer REU. I was able to gain a crucial understanding of the building blocks of machine learning, and this will help me in unimaginable ways in my future career and as AI continues to advance.

The Innerworkings of ChatGPT and Why it Succeeds

Stephen Wolfram's comprehensive exploration of AI-models in “What is ChatGPT Doing... And Why Does it Work?” reveals the fascinating mechanics behind one of the most remarkable AI achievements of our time. At its core, ChatGPT performs a deceptively simple task: predicting the next word in a sequence, one word at a time, based on patterns learned from billions of pages of human-written text. ChatGPT operates by continuously asking "given the text so far, what should the next word be?" For each position, it generates a ranked list of possible words with their associated probabilities. Rather than always selecting the highest-probability word (which produces a flat, repetitive text), ChatGPT introduces *controlled randomness* through a "temperature" parameter. This allows it to occasionally choose lower-ranked words at random, creating more interesting and creative output. The system doesn't actually work with complete words but with "tokens"—language units that might be whole words or fragments like "pre" or "ing." This tokenization approach helps ChatGPT handle rare words, compound terms, and non-English text more effectively, while also enabling it to occasionally generate novel word combinations.

Underneath ChatGPT's conversational abilities lies a massive neural network with 175 billion parameters—weights that determine how the system processes information. A neural network is essentially the idea of a computer trying to function as a *human* brain. The network uses a "transformer" architecture, specifically designed for processing sequential data like text. Unlike earlier neural networks that connected every neuron to every other neuron in adjacent layers, transformers use an "attention" mechanism that allows the system to focus on relevant parts of the input sequence. This attention mechanism is crucial for understanding language

because it enables ChatGPT to connect words that might be separated by many positions in a sentence. For example, it can link a verb to a noun that appeared much earlier, capturing the grammatical relationships that make human language meaningful. ChatGPT learned to generate human-like text by reading massive amounts of text from the internet, books, and other sources. The training process is quite interesting: the system tries to predict the next word in all these texts, and when it gets something wrong, it adjusts those 175 billion parameters slightly to do better next time. What's remarkable is that this training is mostly unsupervised. Nobody had to sit there and tell ChatGPT whether it was a good sentence or not. The system figured out patterns in human language just by trying to predict what comes next, over and over again, millions of times. However, the raw training wasn't enough to make ChatGPT truly useful. The breakthrough came from an additional step where real humans interacted with the system, rating its responses. This human feedback was used to fine-tune ChatGPT's behavior, teaching it not just to generate grammatically correct text, but to be helpful and honest in its interactions.

One of the most intriguing aspects of ChatGPT is how it represents meaning through "embeddings"—arrays of numbers that capture the essence of words, phrases, or entire passages. Words with similar meanings are positioned close to each other in this high-dimensional mathematical space. For example, "dog" and "puppy" would have similar embeddings, while "dog" and "philosophy" would be far apart. These embeddings aren't explicitly programmed but emerge naturally from the training process. The system discovers these meaningful representations by observing how words are used in context across billions of text samples. This creates a "meaning space" in a sense where semantic relationships are preserved as geometric relationships.

Wolfram highlights how the success of ChatGPT suggests something profound about human language that researchers are still trying to fully understand. The fact that a system can generate convincing human-like text just by learning statistical patterns implies that language may possibly be more predictable than we previously thought. This doesn't imply that human communication is mechanical or simple—rather that there might be discoverable principles underlying how we use language that we haven't even fully recognized yet ourselves.

Understanding these principles could lead to both better AI systems and deeper insights into how human communication actually works. I believe ChatGPT represents a remarkable achievement in artificial intelligence, demonstrating that systems with relatively simple architectures can exhibit sophisticated language behaviors when trained on a sufficient amount of data. However, it also reveals the boundaries of our current approaches. The system generates text that sounds human-like but may lack genuine understanding or the ability to perform complex reasoning.

And if we otherwise bypass this hindrance, what does it mean for creative writing as a concept, one that is built on the basis of human emotion; will AI ever be able to indistinguishably produce content that not only rivals but “surpasses” human content in its ability to express emotion? I suppose the key insight from ChatGPT's success is that human language, despite its complexity, contains enough regularity and structure to be captured by large-scale pattern matching. This discovery opens new avenues for both AI development and our understanding of human cognition. As we continue to develop AI systems, ChatGPT's blend of massive scale, clever architecture, and careful training provides a template for future progress, while its limitations remind us of the challenges that remain in creating truly *intelligent* machines.